Unit 3.1 Printables: Changing Earth, Changing Life

Standard: HS-ESS2-7. Construct an argument based on evidence about the simultaneous coevolution of Earth's systems and life on Earth. Targets are labeled ESS2-7.1 to ESS2-7.6.

Everything the class needs for this unit is in this one file, organized by lesson. Print on white paper in black ink only. "Copies" says how many to print: per student, per pair, per group or one for the class. Each item is a dashed box; cut cards along the dashed lines. Sources are cited on any item that uses real data; items marked illustrative use simplified numbers made up for class.

Lesson	Printable	Use	Copies
1	Card sort: 12 events in Earth and life history	Cut apart; pairs sort the cards	One set per pair
1	Timeline strip and E / L / B chart	Glue under "Card sort" in the notebook	One per student
1	Unit CER evidence log	Glue into the back of the notebook; add rows all unit	One per student
2	Cause-and-effect cards	Cut apart; pairs sort the cards	One set per pair
2	Sorting chart and diagrams	Glue under "Sort" in the notebook	One per student
3	Oxygen story: data table and graph grid	Glue under "Graph" in the notebook	One per student
3	Mass extinction data and the rust record	Glue under "Mass extinctions" in the notebook	One per student
3	Build a chain	Glue under "Build a chain" in the notebook	One per student
4	Three case files	Station reading and table; glue under "Case files"	One per student
5	Evidence bank: six cards	Cut apart; add three to the evidence log	One set per student
5	Argument planner and peer-check scale	Glue under "Plan" in the notebook	One per student
6	Evidence Court: eight evidence cards	Cut apart; pairs sort and rate	One set per pair
6	Rating table and argument frames	Glue under "Sort and rate" in the notebook	One per student
7	Earth's systems web: link cards	Cut apart; pairs build the web	One set per pair
7	Earth's systems web: template and reading chart	Glue under "Systems web" in the notebook	One per student
8	The timeline: eight rows	Glue under "The timeline" in the notebook	One per student
9	Review packet A to C	Sections for CFAs under mastery; everyone does Section A	One per student
9	Review packet D to F	Sections for CFAs under mastery; everyone does Section F	One per student

Lessons with no printable (10) use the notebook and the slides only.

UNIT 3.1 · LESSON 1 · CARD SORT

Card sort: 12 events in Earth and life history

Use: Cut apart; pairs sort the cards · Copies: One set per pair

Cut out the 12 cards. For each event decide: did Earth change, did life change, or both? Then place each card on your timeline strip.

A · Earth forms · about 4.6 billion years ago	**B · Oceans of liquid water form · about 4 billion years ago**	**C · First simple cells (microbes) · about 3.5 billion years ago**
D · Cells that use photosynthesis spread · at least 2.7 billion years ago	**E · Oxygen builds up in the air and oceans · starting about 2.4 billion years ago**	**F · First animals · about 600 million years ago**
G · Land plants spread · about 470 million years ago	**H · Continents join into the supercontinent Pangaea · about 300 million years ago**	**I · Largest mass extinction (end of the Permian) · 252 million years ago**
J · Asteroid strike; non-bird dinosaurs go extinct · 66 million years ago	**K · Ice ages come and go · many times; the last ended about 12,000 years ago**	**L · Humans appear · about 300,000 years ago**

Sources: Earth's age, USGS "Age of the Earth" (about 4.54 billion years); oxygen rise, Lyons, Reinhard and Planavsky (2014), Nature 506:307-315; first animals, Narbonne (2005), Annual Review of Earth and Planetary Sciences 33:421-442; land plants, Rubinstein et al. (2010), New Phytologist 188:365-369; end-Permian extinction, Burgess, Bowring and Shen (2014), PNAS 111:3316-3321; end-Cretaceous impact, Renne et al. (2013), Science 339:684-687; earliest Homo sapiens fossils, Hublin et al. (2017), Nature 546:289-292. Dates are rounded; scientists revise them as new evidence arrives.

UNIT 3.1 · LESSON 1 · CARD SORT

Timeline strip and E / L / B chart

Use: Glue under "Card sort" in the notebook · Copies: One per student

Each box is 200 million years. The left end is 4.6 billion years ago and the right end is today. Write each card's letter in the box where it belongs.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23

Box 1 = 4.6 to 4.4 billion years ago. Box 23 = the most recent 200 million years to today.

Card	Event	E, L or B?	Box
A	Earth forms		
B	Oceans of liquid water form		
C	First simple cells (microbes)		
D	Cells that use photosynthesis spread		
E	Oxygen builds up in the air and oceans		
F	First animals		
G	Land plants spread		
H	Continents join into the supercontinent Pangaea		
I	Largest mass extinction (end of the Permian)		
J	Asteroid strike; non-bird dinosaurs go extinct		
K	Ice ages come and go		
L	Humans appear		

Which cards did you mark both? Why could both be true? _____

Where do most of the cards crowd together? What does that say about how fast things changed late in Earth's history?

_____

UNIT 3.1 · LESSON 1 · EVIDENCE LOG

Unit CER evidence log

Use: Glue into the back of the notebook; add rows all unit · Copies: One per student

Add one row each time you find evidence about how Earth and life changed each other. Kind: rock, fossil, data, or model.

Evidence	Kind (rock, fossil, data, model)	What it shows

UNIT 3.1 · LESSON 2 · CAUSE-AND-EFFECT CARDS

Cause-and-effect cards

Use: Cut apart; pairs sort the cards · Copies: One set per pair

For each card write E->L if Earth changed life, or L->E if life changed Earth, then write what changed.

1 · Cells that use photosynthesis release oxygen into the oceans and air.	**2 · Volcanoes throw ash into the sky. The ash blocks sunlight for years, and many plants die.**
3 · Lichens and plant roots break rock into tiny pieces that become soil.	**4 · Sea level drops and a shallow sea dries up. Its animals lose their habitat.**
5 · Corals build reefs that slow waves and protect the shore.	**6 · An ice age spreads cold climate. Some animals evolve thick fur.**
7 · Forests take in carbon dioxide and store carbon in wood, and later in coal.	**8 · An asteroid strikes. Dust darkens the sky, and many species go extinct.**

Examples are standard textbook cases; no data.

Sorting chart and diagrams

Use: Glue under "Sort" in the notebook · Copies: One per student

Card	E->L or L->E	What changed
1		
2		
3		
4		
5		
6		
7		
8		

How many cards are E->L? ____ How many are L->E? ____ Change between Earth and life goes in _____ directions.

Draw your own pair: one E->L card and one L->E card. Two boxes joined by an arrow, labeled (example: Ash cloud -> Plants die).

Diagram 1

Diagram 2

Under each: First, _____ so _____.

Oxygen story: data table and graph grid

Use: Glue under "Graph" in the notebook · Copies: One per student

Graph the oxygen story. Put time on the x-axis (4.0 billion years ago on the left, 0 today on the right). Put oxygen (% of air) on the y-axis from 0 to 30. Plot and connect the points.

Time (billions of years ago)	Oxygen in the air (% of air, simplified)
4.0	0
3.0	0 (a trace)
2.5	0 (a trace)
2.0	2
1.0	3
0.5	15
0.3	30
0 (today)	21

Graph area (draw your axes)

About when does oxygen first rise above a trace? Label that spot Great Oxidation Event. What process made this oxygen?

_____

When is oxygen higher than it is today? What do you think happened to it afterward? _____

Values are simplified from published reconstructions: Lyons, Reinhard and Planavsky (2014), Nature 506:307-315 (rise of oxygen); Berner (2009), Geochimica et Cosmochimica Acta 73:5-23 (Phanerozoic oxygen model). Estimates vary widely; treat the numbers as a teaching model.

Mass extinction data and the rust record

Use: Glue under "Mass extinctions" in the notebook · Copies: One per student

The rust record

Iron dissolved in ancient ocean water. When oxygen from photosynthesis reached it, the iron rusted and sank to the seafloor in layers. These layers are banded iron formations.

Which came first: oxygen from living things, or the rust layers? Explain. _____

Is this an example of life changing Earth, or Earth changing life? Why? _____

Mass extinctions: reading the data

Mass extinction	Time (million years ago)	Species lost (approximate %)
End-Ordovician	445	85
Late Devonian	375	75
End-Permian	252	96
End-Triassic	201	80
End-Cretaceous	66	76

Which extinction killed the greatest percent of species? _____

Many of these were caused by Earth events, such as volcanoes or an asteroid. Explain how an Earth event changes life.

_____

Predict: what happens to the living things that survive an extinction? _____

Source: Barnosky et al. (2011), Nature 471:51-57, "Has the Earth's sixth mass extinction already arrived?" (approximate percent of species lost in each of the "Big Five"; ranges vary by study and are rounded).

UNIT 3.1 · LESSON 3 · BUILD A CHAIN

Build a chain

Use: Glue under "Build a chain" in the notebook · Copies: One per student

Fill in the empty boxes with what happened next. Mark each arrow E->L or L->E.

Cells use photosynthesis	->		->	Iron in the ocean	->	Oxygen builds up in the air	->	An ozone layer forms and

Fill in: box 2 _____; box 3 iron in the ocean _____; box 5 and _____.

Write one sentence: "Because _____ happened, _____ changed, which led to _____."

_____

_____

_____

Three case files

Use: Station reading and table; glue under "Case files" · Copies: One per student

Case A: Oxygen, ozone, and land
Oxygen from photosynthesis rose in the air. Some oxygen became ozone, a layer high in the air that blocks much of the Sun's harmful ultraviolet light. Before the ozone layer, life lived mostly in water. After it, plants and animals could live on land.

Case B: Soil and the first forests
Microbes, lichens, and then plants spread on bare rock about 470 million years ago. They broke rock down and added dead matter, so soil formed. Deep soil let larger plants grow. Roots broke rock further and plants took in carbon dioxide. As forests spread, carbon was buried as coal and the air's carbon dioxide fell, which helped cool the climate.

Case C: The end-Permian mass extinction
About 252 million years ago, huge volcanic eruptions released carbon dioxide and other gases. The climate warmed and the oceans lost oxygen. About 96% of ocean species died. Over the next several million years, survivors spread into empty habitats and new groups, including early dinosaurs, evolved.

Case	Event	Effect on life	Effect back on Earth	Arrow directions
A				
B				
C				

What do all three cases have in common? _____

Sources: Algeo and Scheckler (1998), Phil. Trans. R. Soc. B 353:113-130 (land plants, weathering, and ocean oxygen); Burgess, Bowring and Shen (2014), PNAS 111:3316-3321 (end-Permian timing); percent extinction as in Barnosky et al. (2011), Nature 471:51-57.

Evidence bank: six cards

Use: Cut apart; add three to the evidence log · Copies: One set per student

Read the six pieces of evidence. Add three of them to your evidence log and mark the kind.

E1 · Banded iron formations: layers of rusty iron rock formed about 2.5 to 1.8 billion years ago. · Kind: Rock · Shows: Oxygen reached the oceans and rusted iron.	**E2 · Stromatolites: layered rock mounds made by cyanobacteria, found from 3.5 billion years ago to today. · Kind: Fossil · Shows: Photosynthetic life existed very early.**
E3 · Oxygen data: about 0% before 2.4 billion years ago, about 2% by 2.0 billion, and about 15% by 0.5 billion. · Kind: Data · Shows: Oxygen in the air rose over time.	**E4 · Fossils of the first land plants (about 470 million years old); ancient soils appear after them. · Kind: Fossil and rock · Shows: Plants changed the land and made soil.**
E5 · Five mass extinctions in the last 450 million years; after each, new groups of animals spread within millions of years. · Kind: Data · Shows: Earth events change life, and survivors change the living world.	**E6 · Reef limestone built by corals stretches for hundreds of kilometers; shorelines behind reefs erode less. · Kind: Rock · Shows: Living things change coasts.**

Sources: Bekker et al. (2010), Economic Geology 105:467-508 (banded iron formations); Allwood et al. (2006), Nature 441:714-718 (3.5 billion-year-old stromatolites); Rubinstein et al. (2010), New Phytologist 188:365-369 (early land plants); Barnosky et al. (2011), Nature 471:51-57 (extinctions). Oxygen values are simplified.

UNIT 3.1 · LESSON 5 · PLAN WITH THE FRAMES

Argument planner and peer-check scale

Use: Glue under "Plan" in the notebook · Copies: One per student

Plan with the frames

Claim: Earth and life changed each other because _____.

Evidence 1 (life changed Earth): _____ (use a number or a place).

Evidence 2 (Earth changed life): _____ (use a number or a place).

Evidence 3: _____.

Reasoning: This evidence shows _____ because _____.

Counterclaim: Someone might say _____. That does not fit because _____.

Peer check: score your partner 1 to 4

Score	What it looks like
4	A clear claim that goes both ways. Three pieces of evidence, with numbers, from at least two kinds. Reasoning that explains why each piece supports the claim. A fair counterclaim.
3	A clear claim and two or three pieces of evidence. Reasoning is there but one piece is missing or general.
2	A claim and some evidence, but little reasoning, or the claim goes only one way.
1	A claim with little or no evidence, or a major mistake.

One star (something strong): _____ One step (something to add): _____

Evidence Court: eight evidence cards

Use: Cut apart; pairs sort and rate · Copies: One set per pair

For each card write the kind (rock, fossil, chemical data, model, or story) and a strength score from 1 to 3. Strong evidence is direct (you can touch or measure it), checkable (others can repeat it), and agrees with other kinds of evidence.

C1 · Banded iron formation: layers of rusty rock, about 2.4 billion years old.	**C2 · Measurements of ancient rock chemistry show the oceans and air held very little oxygen before 2.4 billion years ago.**
C3 · Stromatolites: fossil mounds built by photosynthetic microbes, 3.5 billion years old.	**C4 · A computer model that shows oxygen rising when photosynthetic microbes spread.**
C5 · Fossils of the first animals appear after oxygen rose.	**C6 · Ancient soil layers beneath fossil forests.**
C7 · Carbon in rocks has a chemical signature made by living things.	**C8 · A short story in an old book saying "the air was always the same." (No data.)**

Cards summarize evidence discussed in Bekker et al. (2010), Economic Geology 105:467-508; Allwood et al. (2006), Nature 441:714-718; Lyons, Reinhard and Planavsky (2014), Nature 506:307-315.

Rating table and argument frames

Use: Glue under "Sort and rate" in the notebook · Copies: One per student

Card	Kind	Strength (1 to 3) and why
C1		
C2		
C3		
C4		
C5		
C6		
C7		
C8		

Build one argument from several kinds of evidence

Claim: _____. Evidence: a rock _____, a fossil _____, and chemical data or a model _____.

Why they agree: These different kinds of evidence point to the same story because _____.

Skeptic: A skeptic says _____. This evidence answers because _____.

_____

_____

_____

UNIT 3.1 · LESSON 7 · SYSTEMS WEB

Earth's systems web: link cards

Use: Cut apart; pairs build the web · Copies: One set per pair

Draw four boxes in your notebook: Geosphere, Hydrosphere, Atmosphere, Biosphere (one in each corner). Draw an arrow for each link card and label it with a short phrase. Put a star on each arrow where living things changed a non-living system. Add two links of your own.

1 · Volcanoes release gases into the air (geosphere to atmosphere).	**2 · Rain falls and fills the oceans (atmosphere to hydrosphere).**
3 · Photosynthetic cells release oxygen (biosphere to atmosphere).	**4 · Oxygen rusts iron dissolved in the ocean (atmosphere to hydrosphere and geosphere).**
5 · Ozone blocks harmful sunlight so life can live on land (atmosphere to biosphere).	**6 · Plants and roots break rock and make soil (biosphere to geosphere).**
7 · Forests take in carbon dioxide and bury carbon as coal (biosphere to atmosphere and geosphere).	**8 · Ice ages change sea level and where life can live (atmosphere and hydrosphere to biosphere).**
9 · Corals build reefs that change the shore (biosphere to geosphere).	**10 · Mass extinctions change which living things survive (geosphere and atmosphere to biosphere).**

UNIT 3.1 · LESSON 7 · SYSTEMS WEB

Earth's systems web: template and reading chart

Use: Glue under "Systems web" in the notebook · Copies: One per student

Geosphere (rock)	Hydrosphere (water)

Atmosphere (air)	Biosphere (living things)

Reading notes: Lesson 8.1, Earth's Origin (pp. 206-218)

Stage of Earth	Which system changed

Which system has the most arrows pointing out of it? _____ Which has the most pointing in? _____

The timeline: eight rows

Use: Glue under "The timeline" in the notebook · Copies: One per student

Read the eight rows. In the last column write how Earth and life are linked in that row. Circle two rows that form a loop (life changes Earth, then Earth changes life).

Row	When	What happened to Earth	What happened to life	How they are linked
1	3.5 billion years ago	Oceans and shallow seas exist	First cells build layered mounds (stromatolites)	
2	2.7 to 2.4 billion years ago	Iron dissolved in the oceans rusts; the air starts to gain oxygen	Photosynthetic cells spread	
3	2.4 to 1.8 billion years ago	Oxygen rises in oceans and air; rusty iron layers settle on the seafloor	New kinds of cells that use oxygen evolve	
4	600 million years ago	More oxygen in the air and oceans; an ozone layer forms	First animals appear	
5	470 million years ago	The ozone layer blocks most harmful sunlight on land	Plants spread onto land; soils form	
6	350 to 300 million years ago	Forests bury carbon; oxygen is high and carbon dioxide falls	Giant insects and the first large land animals	
7	252 million years ago	Huge volcanic eruptions warm the climate; ocean oxygen drops	About 96% of ocean species go extinct	
8	66 million years ago	An asteroid strike darkens the sky and cools the climate	Non-bird dinosaurs go extinct; mammals diversify	

Sources: Earth's age, USGS "Age of the Earth" (about 4.54 billion years); oxygen rise, Lyons, Reinhard and Planavsky (2014), Nature 506:307-315; first animals, Narbonne (2005), Annual Review of Earth and Planetary Sciences 33:421-442; land plants, Rubinstein et al. (2010), New Phytologist 188:365-369; end-Permian extinction, Burgess, Bowring and Shen (2014), PNAS 111:3316-3321; end-Cretaceous impact, Renne et al. (2013), Science 339:684-687; earliest Homo sapiens fossils, Hublin et al. (2017), Nature 546:289-292. Dates are rounded; scientists revise them as new evidence arrives. Giant insects and carbon burial: Berner (2009).

Write your argument frames

Claim: Earth's systems and life changed together because _____. Counterclaim: Someone says life only reacts to Earth. Use two rows to respond.

UNIT 3.1 · LESSON 9 · REVIEW WORK TIME

Review packet A to C

Use: Sections for CFAs under mastery; everyone does Section A · Copies: One per student

Section A: Vocabulary (everyone)

Word bank: atmosphere, biosphere, geosphere, hydrosphere, photosynthesis, ozone layer, mass extinction, banded iron formation, stromatolite, evidence, argument, feedback

1. Cells use _____ to make sugar and release oxygen. 2. A layer of oxygen high in the air that blocks harmful sunlight is the _____. 3. A time when a very large share of species die out is a _____. 4. Layers of rusty iron rock formed when oxygen reached the oceans: _____. 5. Layered mounds built by microbes and found as fossils: _____. 6. The living things on Earth are the _____; the air is the _____. 7. When a change in one part leads to a change back in the first part, we call it _____. 8. Facts or data that support a claim are _____.

Section B: Earth changed or life changed? (ESS2-7.1)

Write E, L, or B for each: ice ages come and go ___; photosynthetic cells spread ___; Pangaea breaks apart ___; land plants spread ___; oxygen builds up in the air ___; the first animals appear ___.

Name one example of each kind from memory. E: _____ L: _____ B: _____

Section C: Which way, and what next? (ESS2-7.2 and 7.3)

1. Corals build a reef. Which way did the change go? _____ What happens next to the shore? _____

2. A volcano blocks sunlight for years. Which way? _____ Write a chain of three events that follows.

_____

3. Plants spread on land. Name one thing they change in the air _____ and one thing in the soil _____.

Review packet D to F

Use: Sections for CFAs under mastery; everyone does Section F · Copies: One per student

Section D: Kinds of evidence (ESS2-7.4 and 7.5)

Sort these: a banded iron formation; ancient rock chemistry; stromatolite fossils; a computer model; a story with no data. Write the kind of each (rock, fossil, data, model, story) and rank them strongest to weakest. Why does combining kinds make a claim stronger?

_____

_____

_____

_____

Section E: Score it yourself (ESS2-7.4 to 7.6)

Use the four-point scale from Lesson 5. Question: Did life change Earth, or only Earth change life?

Student A: "Life changed Earth because there was a lot of life." Score: ___ Why? _____

Student B: "Life changed Earth. Stromatolite fossils show photosynthetic cells lived 3.5 billion years ago, and the rusty banded iron layers about 2.4 billion years old show oxygen reached the ocean. Rust needs oxygen, so living things that make oxygen must have changed the ocean and air. Oxygen then rose to about 15% of the air by 0.5 billion years ago, which let animals evolve. Someone might say volcanoes made the oxygen, but the fossils and layers point to living things." Score: ___ Why? _____

Student C: "Oxygen went up, so life changed Earth." Score: ___ Why? _____

Rewrite Student C's answer so it earns a 4:

_____

_____

_____

Section F: Write It (ESS2-7.6)

One paragraph (8 to 10 sentences) arguing that Earth's systems and life changed together. Use at least three pieces of evidence from at least two kinds, one loop, and one counterclaim.

_____

_____

_____

_____